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**BSIT**

**1-MONTH NETWORK FORENSICS (FEBRUARY)**

***WEEK 2 – Protocol Analysis & Suspicious Network Activity***

### ***Overview***

Week 2 focuses on analyzing common network protocols and identifying suspicious or malicious behavior within network traffic. While Week 1 introduced packet structures and basic traffic analysis, this week emphasizes detecting abnormal activity through protocol-level inspection.

Using the network forensic tool Wireshark, investigators analyze captured traffic to identify suspicious patterns such as unauthorized access attempts, malicious DNS queries, data exfiltration, and network scanning activities.

By the end of Week 2, the learner should be able to identify abnormal traffic patterns, reconstruct sessions, and extract files from packet captures.

## ***1. HTTP and HTTPS Forensic Analysis***

### **HTTP Traffic Analysis**

HTTP (Hypertext Transfer Protocol) is widely used for communication between web clients and servers. Because HTTP traffic is often transmitted in plain text, it provides valuable information during forensic investigations.

Common elements analyzed in HTTP traffic include:

- Request methods (GET, POST, PUT)
- Host headers
- User-Agent strings
- Cookies, URLs, and parameters
- Response status codes.

### ***Example HTTP Request Header***

*GET /index.html HTTP/1.1*

*Host: example.com*

*User-Agent: Mozilla/5.0*

*Accept: text/html*

From a forensic perspective, investigators analyze HTTP traffic to detect:

- Suspicious downloads
- Unauthorized file uploads
- Data exfiltration

- Malware communication with command-and-control servers

Using Wireshark, analysts can filter HTTP traffic using:

`http`

### **HTTPS Traffic Analysis**

HTTPS (Hypertext Transfer Protocol Secure) encrypts communication using SSL/TLS encryption. Although the payload is encrypted, investigators can still analyze metadata such as:

- Source and destination IP addresses
- Server Name Indication (SNI)
- TLS handshake information
- Certificate details
- traffic patterns

HTTPS forensic analysis can help identify:

- Suspicious encrypted connections
- Malware contacting external servers
- Unusual TLS certificate behavior

**Example filter:**

`tls`

## **2. DNS Traffic Analysis and Abuse Techniques**

### **DNS Overview**

DNS (Domain Name System) translates domain names into IP addresses.

**Example:**

`example.com` → `12.123.154.32`

DNS queries are extremely valuable in forensic investigations because they reveal which domains a device attempts to access.

### **DNS Forensic Analysis**

Investigators analyze DNS traffic to detect:

- Communication with malicious domains
- Domain Generation Algorithms (DGA)
- Data exfiltration through DNS tunneling

Example filter in Wireshark:

`dns`

Common DNS fields analyzed include:

- Query name
- Query type
- Response codes
- TLL values

## **DNS Abuse Techniques**

### **DNS Tunneling**

Attackers may encode data inside DNS queries to bypass security controls.

Example suspicious domain:

`data.exfiltration.malicious-domain.com`

### **Domain Generation Algorithms (DGA)**

Some malware generates random domain names to communicate with attackers.

***Example suspicious domain:***

`xjsklqwe.biz`

Indicators of DNS abuse include:

- Unusually long domain names
- Large numbers of failed DNS queries
- High frequency of DNS request

## **3. FTP, SMTP, and SMB Traffic Basics**

Understanding additional protocols is important in network forensic investigations.

### **FTP (File Transfer Protocol)**

FTP is used for transferring files between systems.

Common FTP commands include:

- USER
- PASS
- LIST
- RETR
- STOR

Because FTP often transmits credentials in plain text, investigators may recover:

- Usernames
- Passwords
- Transferred files

Example filter in Wireshark:

ftp

### **SMTP (Simple Mail Transfer Protocol)**

SMTP is used to send emails across networks.

Investigators analyze SMTP traffic to detect:

- Spam campaigns
- Phishing emails
- Malware attachments
- Email spoofing

#### ***Example filter:***

*smtp*

SMTP headers may reveal:

- Sender address
- Recipient address
- Email relay path
- Message ID

### **SMB (Server Message Block)**

SMB is used for file sharing and resource access in Windows environments.

SMB forensic analysis helps detect:

- Unauthorized file access
- Lateral movement inside networks
- Data exfiltration

#### ***Example filter:***

*smb*

## ***4. Indicators of Suspicious Network Activity***

During investigations, analysts look for **Indicators of Compromise (IOCs)**.

### **Unusual Ports and Protocols**

Certain ports are typically associated with specific services.

#### **Port Service**

80 HTTP

443 HTTPS

## Port Service

21 FTP

25 SMTP

Suspicious indicators include:

- HTTP traffic on non-standard ports
- Unknown protocols on unusual ports
- External communication on high-numbered ports

## Beaconing Patterns

Beaconing occurs when malware periodically contacts a command-and-control server.

Characteristics include:

- Regular connection intervals
- Small repeated packets
- Consistent destination IP

Beaconing may indicate botnet or malware communication.

## Repeated Failed Connections

Repeated connection failures may indicate:

- Brute-force login attempts
- Network scanning
- Service discovery

Indicators include:

- Multiple login failures
- Repeated authentication attempts
- High number of connection attempts

## 5. Traffic Filtering and Stream Reconstruction

Network forensic tools allow analysts to isolate traffic for analysis.

Using display filters in Wireshark:

### ***Filter DNS traffic:***

*dns*

### ***Filter traffic from a specific IP:***

*ip.addr == 192.168.1.10*

### ***Filter TCP traffic:***

*tcp*

## **Stream Reconstruction**

Stream reconstruction allows analysts to rebuild full network conversations.

Steps:

1. Select a packet in Wireshark
2. Right-click the packet
3. Select **Follow** → **TCP Stream**

This displays the entire communication between the client and server and may reveal:

- COMMANDS
- CREDENTIALS
- TRANSFERRED DATA

## **6. File Extraction from Network Traffic**

Files transferred over a network can often be extracted from packet captures.

Using Wireshark, analysts can export files transferred through protocols such as:

- HTTP
- FTP
- SMB

Steps to extract files:

1. Open the PCAP file
2. Navigate to **File** → **Export Objects**
3. Select the protocol type (HTTP, SMB, etc.)
4. Save the extracted file

This technique helps investigators recover:

- Malware samples
- Downloaded documents
- Transferred images
- Suspicious executables

## **7. Common Attack Patterns**

### **Network Scanning**

Attackers scan networks to discover open ports and vulnerable services.

Indicators include:

- Sequential port connection attempts
- Multiple SYN packets
- Rapid connection attempts to different ports

***Example scanning behavior:***

*192.168.1.5 → Ports 20, 21, 22, 23, 25, 80*

## **Brute-Force Attacks**

Brute-force attacks attempt to guess passwords by repeatedly trying login credentials.

Indicators include:

- Repeated authentication attempts
- Failed login messages
- Rapid connection attempts to login services

Common targets include:

- FTP
- SSH
- SMTP
- Web login pages

## **Unauthorized Data Transfer**

Data exfiltration occurs when sensitive data is transferred outside the network.

Indicators include:

- Large files
- Unexpected outbound traffic
- Connections to unknown external servers

Protocols often used in data exfiltration:

- HTTPD
- FTP
- DNS tunneling

## **8. Practical Limitation**

The practical exercises for Week 2 were intended to be conducted using Wireshark to analyze packet capture (PCAP) files and observe real network traffic.

However, the software was not functioning properly on the system used during the training period, which prevented the opening and analysis of packet capture files. Due to this technical issue, the practical packet analysis activities could not be performed.

Despite this limitation, the procedures and methodologies for protocol analysis were reviewed and documented to demonstrate the standard workflow used in network forensic investigations.

## **9. Conclusion**

Week 2 provided an understanding of how network protocols can be analyzed to detect suspicious and malicious activities within network traffic. By examining protocols such as HTTP, DNS, FTP, SMTP, and SMB, investigators can identify indicators of compromise and abnormal communication patterns.

Although the practical packet analysis could not be completed due to technical issues with the analysis tool, studying the investigation procedures still provided valuable knowledge about how network forensic investigations are conducted.

## **10. Reflection**

During Week 2, I learned how different network protocols such as HTTP, DNS, FTP, SMTP, and SMB can be analyzed to detect suspicious or malicious activities within network traffic. Understanding how these protocols operate helps investigators identify abnormal behavior, unauthorized access attempts, and potential data exfiltration.

Although the practical packet analysis could not be performed due to technical issues with Wireshark, studying the investigation procedures still improved my understanding of how network forensic analysis is conducted. I learned how analysts filter traffic, reconstruct sessions, and extract files from packet captures to identify security threats.

Overall, this week helped strengthen my knowledge of protocol analysis and the importance of monitoring network traffic in cybersecurity investigations. These skills are essential for detecting attacks and responding to security incidents in real-world network environments.